



SASTRA
ENGINEERING • MANAGEMENT • LAW • SCIENCES • HUMANITIES • EDUCATION
DEEMED TO BE UNIVERSITY
(U/S 3 of the UGC Act, 1956)



THINK MERIT | THINK TRANSPARENCY | THINK SASTRA

Srinivasa Ramanujan Centre

Department of Electronics and Communication Engineering

Programme: B.Tech - Electronics and Communication Engineering

Course Code & Title: ECE300 – Mini Project

Semester and Year: VI semester and Third year

Batch No: 42

Title of the Project: LeafGuard AI: Predictive Modeling of Plant Stress Using Multisensor Data.

Team Members:

1. 227004026: Galam.Poojitha
2. 227004039: Kapaka.JahnaviKrishna

Supervisor Name:

Dr.A.Subramanian

Base paper details:

Authors: Divisha Garg, Harpreet Singh, Yosi Shacham-Diamand

Paper Title: AdapTree: Data-Driven Approach to Assessing Plant Stress Through the AISensor Synergy

Journal Name: Sensors

Publisher: MDPI, Basel, Switzerland

Volume: 25

Year: 2025

ABSTRACT

Plants are highly vulnerable to environmental stresses caused by heat, drought, and humidity fluctuations, which significantly reduce growth and yield. This project focuses on minimizing plant stress by combining IoT technology with machine learning (ML). Different types of Environmental sensors temperature, relative humidity (RH), vapor pressure deficit (VPD), volumetric water content (VWC), plant weight, light intensity, soil nutrient sensor. The AdapTree algorithm proposed enhance predictive accuracy. Regression-based training with metrics Root mean square error (RMSE), and Mean absolute error (MAE) applied to train the ML model. The AdapTree algorithm provides a reliable, scalable, and effective solution for plant stress detection.

Keywords: Plant stress detection, Temperature, Humidity, vapour pressure deficit (VPD), volumetric water content (VWC), AdapTree algorithm, Machine learning.

Signature of the Supervisor: